

TTIC 31220: Unsupervised Learning and Data Analysis

Homework 4: Density Estimation & Data Analysis

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Introduction

This homework assignment includes a mix of on-paper work and data-based exercises, mainly focusing on density estimation.

1 Feedback (4 points)

This part is intended to give us some useful feedback on the homework.

1. **Collaboration:** Did you collaborate with anyone on this homework? If so, who? - NO
2. **Time:** How long did you spend on this homework? - this one was shorter than usual, so like maybe 4-6 hours total
3. **LLM use:** Did you use any LLM? If so, which one(s) and how? Did it help your learning of the material? - I used Claude to proofread grammar. I wrote all solutions myself; the LLM did not produce solutions for me. I did ask Gemini to help my understanding of the limiting argument in 2.1.
4. **Comments:** Any comments or suggestions that might help us improve the homework or preparation for it (e.g. anything that was unclear, any readings you like better than the assigned ones)?

2 Paper Exercises (49 points)

2.1 Relationship between Gaussian Mixtures and k-means Clustering (24 points)

In this exercise you will show that k -means clustering is equivalent to a limiting case of Gaussian mixture models. Consider a Gaussian mixture with k components over the observed variable $\mathbf{x} \in \mathbb{R}^d$. Let $z \in \{1, 2, \dots, k\}$ be the hidden variable representing the Gaussian component. Then the component densities of the Gaussian mixture are given by

$$p(\mathbf{x}|z = i) = \frac{1}{\sqrt{(2\pi)^d |\Sigma_i|}} e^{-\frac{1}{2}(\mathbf{x} - \mu_i)^T \Sigma_i^{-1} (\mathbf{x} - \mu_i)},$$

where μ_i, Σ_i are the mean vector and covariance matrix, respectively, of the i^{th} component.

1. (16 points) Show that if, for all i , $\Sigma_i = \sigma^2 I$, where σ is a scalar constant I is an identity matrix of the appropriate size, then:

$$\lim_{\sigma^2 \rightarrow 0} p(z = i|x) = \begin{cases} 1 & \text{if } i = \arg \min_j \|\mathbf{x} - \mu_j\| \\ 0 & \text{otherwise} \end{cases}$$

SOLUTION: With $\Sigma_i = \sigma^2 I$ we have $|\Sigma_i| = \sigma^{2d}$ and $\Sigma_i^{-1} = \frac{1}{\sigma^2} I$, so the component density simplifies to

$$p(\mathbf{x} | z = i) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{\|\mathbf{x} - \mu_i\|^2}{2\sigma^2}\right).$$

If we write the posterior via Bayes' rule:

$$p(z = i | \mathbf{x}) = \frac{p(z = i) p(\mathbf{x} | z = i)}{\sum_{j=1}^k p(z = j) p(\mathbf{x} | z = j)}.$$

We can cancel the common factor from the numerator and the denominator, giving

$$p(z = i | \mathbf{x}) = \frac{\pi_i \exp\left(-\frac{\|\mathbf{x} - \mu_i\|^2}{2\sigma^2}\right)}{\sum_{j=1}^k \pi_j \exp\left(-\frac{\|\mathbf{x} - \mu_j\|^2}{2\sigma^2}\right)}, \quad (1)$$

where $\pi_i = p(z = i) > 0$ are the mixing weights.

Then, taking $\sigma^2 \rightarrow 0$. Let $i^* = \arg \min_j \|\mathbf{x} - \mu_j\|$. Divide numerator and denominator of the posterior equation by $\exp(-\|\mathbf{x} - \mu_{i^*}\|^2/2\sigma^2)$:

$$p(z = i | \mathbf{x}) = \frac{\pi_i \exp\left(-\frac{\|\mathbf{x} - \mu_i\|^2 - \|\mathbf{x} - \mu_{i^*}\|^2}{2\sigma^2}\right)}{\sum_{j=1}^k \pi_j \exp\left(-\frac{\|\mathbf{x} - \mu_j\|^2 - \|\mathbf{x} - \mu_{i^*}\|^2}{2\sigma^2}\right)}.$$

For $j \neq i^*$: by definition of i^* , the exponent satisfies $\|\mathbf{x} - \mu_j\|^2 - \|\mathbf{x} - \mu_{i^*}\|^2 > 0$, so

$$\exp\left(-\frac{\|\mathbf{x} - \mu_j\|^2 - \|\mathbf{x} - \mu_{i^*}\|^2}{2\sigma^2}\right) \xrightarrow{\sigma^2 \rightarrow 0} 0.$$

For $j = i^*$: the exponent is 0, so the exponential equals 1 for all σ^2 .

Therefore:

- If $i = i^*$: numerator $\rightarrow \pi_{i^*} \cdot 1$, denominator $\rightarrow \pi_{i^*} \cdot 1$, so $p(z = i^* | \mathbf{x}) \rightarrow 1$. ✓
- If $i \neq i^*$: numerator $\rightarrow \pi_i \cdot 0 = 0$, denominator $\rightarrow \pi_{i^*} > 0$, so $p(z = i | \mathbf{x}) \rightarrow 0$. ✓

So

$$\lim_{\sigma^2 \rightarrow 0} p(z = i | \mathbf{x}) = \begin{cases} 1 & \text{if } i = \arg \min_j \|\mathbf{x} - \mu_j\| \\ 0 & \text{otherwise.} \end{cases} \quad \blacksquare$$

2. (8 points) Explain how this fact makes k -means a limiting case of Gaussian mixtures.

SOLUTION The EM algorithm for a Gaussian mixture iterates between two steps: the e step and the m step. The e step computes soft responsibilities for every data point and the m state updates the means.

Part 1 shows that in the limit $\sigma^2 \rightarrow 0$ with equal covariances $\Sigma_i = \sigma^2 I$, the soft responsibilities degenerate into harder assignments

$$r_{ni} \longrightarrow \mathbf{1} \left[i = \arg \min_j \|\mathbf{x}_n - \mu_j\| \right].$$

Each data point is assigned to its nearest centroid (responsibility 1) and all other components receive responsibility 0. Substituting into the M-step mean update:

$$\mu_i = \frac{\sum_n \mathbf{1}[i_n^* = i] \mathbf{x}_n}{\sum_n \mathbf{1}[i_n^* = i]},$$

which is just the k -means centroid update because it is the mean of all points assigned to cluster i . So, the E-step hard assignment corresponds exactly to the k -means assignment step

Therefore k -means is recovered as the $\sigma^2 \rightarrow 0$ limit of EM for isotropic Gaussian mixtures: the smooth probabilistic responsibilities of the GMM collapse into distance-based assignments of k -means, and the weighted within-cluster variance/obj func becomes the standard k -means sum-of-squared-distances objective.

2.2 HMMs (25 points)

Consider a data set consisting of the following (single) string of 160 symbols from the alphabet $\{A, B, C\}$:

AABBBACABBBACAAAAAAAAAABBBACAAAAABACAAAAAABBBBACAAAAAAAAAAAAABACABACAA
BBACAAABBBBACAAABACAAAABACAABACAAABBBACAAAAABBBBACABBACAAAAABACABACAA
ABACAABBBACAAAABACABBACA

Study this string carefully. Then, describe a hidden Markov model for it. Your description should include the number of states in the HMM, the transition matrix including the values of the elements of the matrix, the emission matrix including the values of its elements, and the initial state probabilities. Provide some description/justification for how you arrived at these numbers. We are not expecting you to code or run the HMM algorithms, but rather to arrive at the answers by careful examination of the data.

SOLUTION: The string has length = 160. The letter C appears exactly 21 times and is immediately preceded by BA — that is, C appears only as part of the three-character substring BAC occurs exactly 21 times. Splitting the string on BAC yields 21 chunks which each consist of a run of A's followed by an optional run of B's:

$$\underbrace{A^{a_1} B^{b_1}}_{\text{chunk 1}} \text{BAC} \underbrace{A^{a_2} B^{b_2}}_{\text{chunk 2}} \text{BAC} \cdots \underbrace{A^{a_{21}} B^{b_{21}}}_{\text{chunk 21}} \text{BAC}$$

- A-run lengths: $\{2, 1, 9, 5, 6, 12, 1, 2, 3, 3, 4, 2, 3, 4, 1, 6, 1, 3, 2, 4, 1\}$; total = 75, mean is approx 3.57.
- B-run lengths in B-chunks: total B-emissions = 21.
- Within every chunk, all A's precede all B's (no A ever follows a B within a chunk).
- 11 of 21 chunks have a non-empty B-run; 10 have $b_i = 0$.

Making an HMM with 5 states

State	Role	next
q_1	A-run within a chunk	A
q_2	B-run within a chunk	B
q_3	First symbol of BAC delimiter	B
q_4	Second symbol of BAC delimiter	A
q_5	Third symbol of BAC delimiter	C

All emissions are deterministic, so the emission matrix becomes:

	A	B	C
q_1	1	0	0
q_2	0	1	0
q_3	0	1	0
q_4	1	0	0
q_5	0	0	1

q_2 and q_3 both emit B which is why we need hidden states. They will only really be distinguishable by their transitions.

Transition matrix (row = from-state, column = to-state):

From \ To	q_1	q_2	q_3	q_4	q_5
q_1	≈ 0.720	≈ 0.147	≈ 0.133	0	0
q_2	0	≈ 0.476	≈ 0.524	0	0
q_3	0	0	0	1.000	0
q_4	0	0	0	0	1.000
q_5	1.000	0	0	0	0

Init state distribution: $\pi = (1, 0, 0, 0, 0)$ — the string begins with A's, so we start in state q_1 .

Transition probabilities scratchwork

- $P(q_1 \rightarrow q_1)$: 75 total A-emissions and 21 departures from q_1 (one per chunk-end), giving $75 - 21 = 54$ self-loops. $P(q_1 \rightarrow q_1) = 54/75$.
- $P(q_1 \rightarrow q_2)$: of 21 departures, 11 enter a B-run. $P(q_1 \rightarrow q_2) = 11/75$.
- $P(q_1 \rightarrow q_3)$: the remaining 10 skip the B-run and go straight to delimiter. $P(q_1 \rightarrow q_3) = 10/75$.
- $P(q_2 \rightarrow q_2)$: 21 total B-emissions in B-runs and 11 departures, giving $21 - 11 = 10$ self-loops. $P(q_2 \rightarrow q_2) = 10/21$.
- $P(q_2 \rightarrow q_3) = 11/21$.
- $q_3 \rightarrow q_4$, $q_4 \rightarrow q_5$, $q_5 \rightarrow q_1$ are all deterministic (probability 1), encoding the fixed BAC delimiter.